ECE-UY 3613 2024 Spring: Sample Midterm Solutions

- **Write your answers CLEARLY and INTELLIGIBLY on the exam book;**
- **Write you NAME and ID on the exam book;**

1) (*12 points*) Given an IP address $128.135.174.79$ and the subnet mask $255.255.224.0$, find:
   a) (*3 points*) the class of this address (Class A, B, C, D or E),
   b) (*3 points*) the subnet part of this address,
   c) (*3 points*) the host part of this address,
   d) (*3 points*) the number of hosts that can be connected in this subnet.

   Answer: Binary IP address is: 1000 0000. 1000 0111. 1010 1110. 0100 1111 Subnet Mask: 1111 1111. 1111 1111. 1110 0000. 0000 0000
   a) Class B (first two bits of IP address is 10)
   b) Subnet part: 1000 0000. 1000 0111. 101, or the corresponding decimal number
   c) Host part: 0 1110 0100 1111, or (3663 decimal)
   d) number of hosts $2^{13} - 2 = 8190$

2) (*12 points*) A large IP packet has a total size of $3,000$ bytes, and its ID is 1000. The packet is to be sent over a link with MTU of $1,120$ bytes, assume the IP header is 20 bytes, please show the following fields for the packets actually transmitted on this link: **length, ID, fragment flag, offset**
   Answer: a) 3000 bytes of total packet, 2980 bytes of data payload,
   With MTU=1120 bytes, each frame has at most $1100 = 137.5 * 8$ bytes of data, Since it must be a multiple of 8, so each frame can have at most 137*8=1096 bytes of data,
   Three frames will be generated:
   Lengths are: 1,116; 1,116, 808;
   IDs: are all 1000
   Flags are: 1, 1, 0
   Offsets: 0, 137, 274

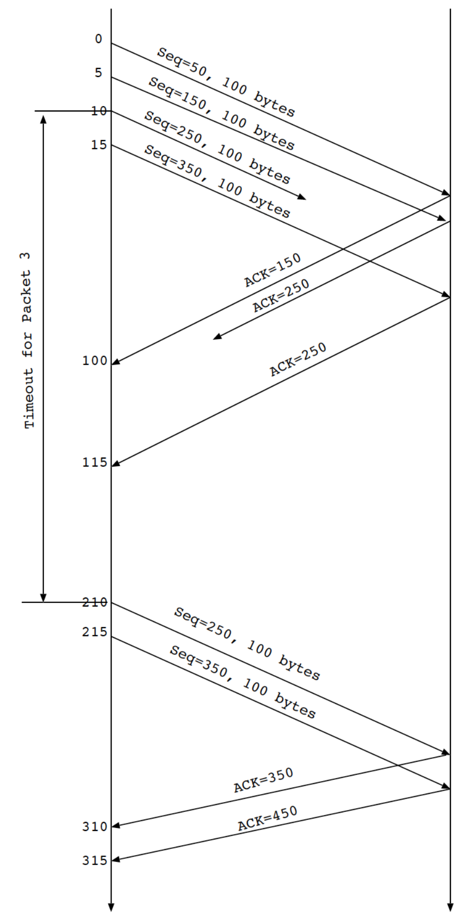
3) (*12 points*) A TCP connection will experience a triple-duplicate-ack packet loss whenever the sender's congestion window reaches 64 segments, and suppose there is no other packet loss in the network, if the round-trip-time of this connection is 100 millisecond, and the packet size is $1,000$ bytes
   a) (*6 points*) how often does this TCP connection experience packet loss? In other words, what is the time interval between two consecutive packet losses?
   b) (*6 points*) what is the average sending rate (measured in bits-per-second) of the TCP sender in long run?

   Answer:
   a) after each loss, TCP will do fast-recovery, reduce its window to 32 segments, then increase it by one segment each round-trip time, until it reaches 64 again, the total time needs is 32 RTTs= 3200 ms. (It is ok if someone use 33 RTTs)
   b) average window size is $(32 + 64)/2 = 48$, the average sending rate:

$$\frac{48 * 1000 * 8}{0.1} = 3.84Mbps$$

4) (*12 points*) At time zero, a TCP sender begins to send out a total of four segments to a receiver, one segment every 5 ms, and each segment contains 100 bytes of data, and the initial sequence number on the sender is 50. Suppose the receiver sends back one acknowledgement for each received segment. Assume segment 3 is lost and the acknowledgement for segment 2 is lost, the one-way delay between the sender and receiver is 50 ms, and the time-out value is 200ms, you can ignore the processing delay on the receiver side. Plot the packet exchanges between the sender and receiver over time.

Answer:



Here we assume the receiver discards out-of-order segments. It is also fine if you assume the receiver keeps out-of-order segment, then the sender will only retransmit segment 250 after time out, when receiver receives it, it will send back ACK=450.

5) (*10 points*) When two users behind the same NAT router access the same website on a web server, suppose the NAT router only has one public IP address, explain how the web server can distinguish requests from the two users, and how the web server's responses are sent back to different users.

Answer: From the web server's point of view, the two users' requests will carry the same public IP address, but from different source port numbers (assigned by the NAT router). The web server's responses will carry different destination port numbers, one for each client. The NAT router will forward the responses to the corresponding clients by looking up its NAT table.

6) (*12 points*) Suppose Alice uses a Web-based e-mail account (such as Gmail or Hotmail) to send out a message to Bob, and the IP address of her email server is initially unknown.
   a) (4 points) What application-layer protocols are needed on Alice's machine for her to send an email?
   b) (4 points) What transport layer protocols are used for each application protocol, and why?
   c) (4 points) Does Alice's email client need to get the IP address of Bob's email server? Briefly explain why or why not.

   Answer:
   a) DNS, HTTP;
   b) DNS uses UDP for low overhead/delay HTTP uses TCP for reliability
   c) No, Alice's email server is responsible for contacting Bob's email server.

7) (*12 points*) For multimedia networking,
   a) (*6 points*) please rank order (from low to high) the following applications: Netflix movie streaming, Skype video calls, and ESPN live sports broadcast, based on their streaming buffer lengths;
   b) (*6 points*) please list two major advantages of using CDN in multimedia networking.

   Answer:
   a) From low to high: Skype, ESPN, Netflix;
   b) (Any two out of the list, or any other reasonable advantages)
   offload the original server;
   push content closer to client, reduce their downloading time;
   reduce traffic on the network;
   reduce the chance of single-point-of failure